NYU SPS X Pulse Foundry

Hackathon CFA

TL;DR

This Call for Applications (CFA) invites graduate students from across NYU to apply for a one-day hackathon with Pulse Foundry, an AI consultancy, incubation lab, and builder community based in New York City. Selected participants will work in teams to tackle a real-world challenge, develop a working solution, and present what they've built to a panel of judges. If you're a strong builder, expect an afternoon of tackling a real-world challenge, making something tangible, and working alongside people who like to build.

About Pulse Foundry

Pulse Foundry is an AI consultancy and incubation lab. On the consulting side, we embed small teams of engineers, data scientists, and solution architects in mid-market companies and help them ship AI systems that run in production, including automations, internal tools, pricing and search work, and more. On the lab side, we take promising ideas and early teams and help turn them into real products. The community grew out of that. We kept meeting sharp people who wanted somewhere to build and compare notes, so we started running events for them. It's now one of the city's larger technical communities.

Community Impact

Most of what Pulse Foundry does is hands-on. They run hackathons where teams come together to build and ship something in an afternoon, along with smaller events, including talks, demos, and working sessions. People come to work on real problems and connect with others doing the same. They also pay attention to who shows up and what they build. Strong work at their events has led to roles, contract opportunities, and professional introductions. That's not the primary purpose of Pulse Foundry, but it's an important part of what can come from

I'm Interested! What's The Challenge?

Search is changing. Instead of typing "best Italian restaurant NYU" into Google and scrolling through ten blue links, people now ask ChatGPT, Gemini, Perplexity, and Claude directly, expecting a confident, specific answer. That shift creates a new, mostly invisible battlefield: **AI Search Optimization (AISO)**. Traditional SEO tells you how to rank on Google. Almost nothing tells a small business how to get recommended when an AI assistant is asked: *"Where should I eat?"*

Your challenge is to **build an agent that helps a real restaurant win that battle.**

Your Problem Statement

Given a restaurant's name, website, or location, build an agent that can answer five questions for the owner:

1. **Where do I currently rank in AI search?**
2. **Why are competitors beating me?**
3. **What should I change?**
4. **Can the agent make those changes for me?**
5. **Did my visibility improve?**

This maps to a four-stage loop: **Discovery → Analysis → Action → Measurement.**

What You'll Build

1. Discovery — Simulate the customer

- Generate a realistic bank of queries a customer might actually ask an AI assistant (e.g., "best healthy lunch near NYU," "quiet date-night spot in Greenwich Village," "cheap eats for a study group"). Vary intent, distance, price, cuisine, and occasion.
- Run these queries against one or more AI platforms (via API where available, e.g., Claude, GPT, Gemini, or via structured prompting that simulates how a retrieval-augmented assistant would answer).
- Log, for every query: was the target restaurant mentioned? At what position? Which competitors appeared, and how often?

2. Analysis — Explain the gap

- Build a visibility scorecard: mention rate, average rank, competitor share-of-voice, for a given restaurant across the query set.
- Analyze the *inputs* AI models likely draw on: the restaurant's own website copy, structured data (schema.org/JSON-LD), Google Business Profile, Yelp/TripAdvisor listings, press coverage, blog mentions, and Reddit/forum discussion.
- Diagnose *why* competitors win, e.g., richer menu descriptions, more third-party reviews, dedicated blog content answering "best of" questions, structured FAQ data, more recent press mentions.
- Surface concrete, prioritized gaps (not just a single "AI SEO score").

3. Action — Close the gap

- Generate AI-search-friendly content: optimized "About" copy, menu descriptions, FAQ pages, schema markup (Restaurant, Menu, FAQPage, Review), and locally-relevant landing content (e.g., "Best lunch spots near NYU" written to plausibly earn a citation).
- **Bonus points**: don't just recommend, *act*. E.g., draft and (with permission) publish updated website copy, submit/update structured data, draft outreach emails to local blogs or press, or auto-generate content ready to paste into a CMS.

4. Measurement — Prove it worked

- Re-run the same query bank on a schedule (daily/weekly) and track mention rate and rank over time.
- Present before/after visibility trends so the restaurant can see impact, not just effort.

Sample Case: Greenwich Kitchen

A fictional healthy-food spot near NYU. The agent runs 20 variations of "best healthy lunch near NYU" and finds Greenwich Kitchen is recommended in only 2 of 20 answers, while three competitors dominate.

The agent should be able to say something like:

> *"Competitor A appears in 15/20 answers. It has 340 Google reviews (you have 40), a dedicated 'healthy lunch NYU' blog post that ranks and gets cited, and structured menu data with dietary tags (vegan, gluten-free) that AI models can parse directly. Recommended actions: (1) add schema.org Menu markup with dietary tags, (2) publish an FAQ page answering 'healthiest lunch options near NYU,' (3) run a review-generation campaign, target 100+ reviews in 60 days, (4) pitch a local student-focused blog for a mention."*

That's the bar: specific, evidence-based, and actionable, not a generic score.

Technical Building Blocks (*suggested, not required*)

Layer	Ideas
Query generation	LLM-generated query variants across intent/cuisine/occasion/location; can be templated + expanded
AI platform testing	Claude/GPT/Gemini APIs; Perplexity API; where no API exists, simulate via RAG-style prompting with real web context
Web & data analysis	Website scraping, schema.org/JSON-LD parsing, Google Business Profile & Yelp data, Reddit/forum search, press mention search
Competitive diagnostics	Structured comparison of target vs. top-appearing competitors across content, reviews, structured data, and mentions

Layer	Ideas
Content generation	LLM-generated FAQ, menu copy, schema markup, and locally-relevant blog content
Action layer (bonus)	CMS/webhook integration, auto-generated PRs to a website repo, email drafts, scheduled re-testing
Measurement	Time-series tracking of mention rate/rank per query bank; simple dashboard

Deliverables

- **Working demo** using at least one real restaurant (a real NYU-area spot is ideal, but a realistic mock is fine).
- **A visibility report** showing current AI-search performance vs. named competitors, with specific reasons for the gap.
- **Generated artifacts** the restaurant could actually use, e.g., website copy, FAQ content, structured data snippets.
- **A measurement mechanism** — even a simple "re-run this and compare" loop counts.
- *(Bonus)* Evidence of the agent taking a real action autonomously (publishing content, submitting data, sending outreach, etc.), with human-in-the-loop confirmation where appropriate.

Judging Criteria

Criterion	What we're looking for
Discovery depth	Realistic, varied query simulation across real AI platforms
Analytical rigor	Clear, evidence-based explanation of *why* competitors win, not just a score
Actionability	Specific, usable recommendations and generated content
Autonomy (bonus)	Agent takes real action, not just advice
Measurability	A credible way to track improvement over time

Criterion	What we're looking for
Polish & demo quality	Can you show a judge, in 3 minutes, "here's what's broken, here's what we fixed, here's proof it worked"?

Constraints & Notes for Teams

- You don't need access to every AI platform's API. Build for the ones you can reach, and clearly explain how the approach generalizes to others.
- Be thoughtful about scraping etiquette and rate limits when pulling review/listing data.
- If your agent takes autonomous actions (publishing, emailing), build in a clear human-approval step for the demo. Reversible, transparent actions win more trust than a black box.
- A great submission tells a **before** → **diagnosis** → **fix** → **after** story, not just a technical pipeline.

Think of what you're building as an AI growth agent for the new era of search — not a report generator, but a system that keeps working for the restaurant after the hackathon ends.

Great! So, When Is This Happening?

Date: Friday, October 23, 2026
Time: 9:00 AM – 7:00 PM
Location: NYU SPS MTC | Rooms 1026 & 1027

Hackathon Timeline

This one-day hackathon is open to NYU students, with approximately 100 participants selected at random to form roughly 20 teams. Attendance at the on-site hackathon is required, and all team members must be available to participate for the full event. Teams unable to meet the required dates and milestones outlined below may be ineligible to participate.

- **Hackathon Announced & Application Opens:** Thursday, September 10
- **Application Closes:** Sunday**,** October 11, at 11:59 PM EST
- **Participants Announced:** Friday, October 16
- **Hackathon Event:** Friday**,** October 23 from 9:00 AM to 7:00 PM
    - 9:00 - 9:30 Arrival
    - NYU SPS MTC | Rooms 1026 & 1027
    - 6:30 - 7:00 Concluding & Networking

How Do I Apply?

To apply for the NYU SPS X Pulse Foundry One-Day Hackathon, please submit your application to the Google Form linked below. To be considered, complete the registration form as soon as

possible. Given the level of interest in this event, registered students are expected to commit to attending the full session. Please register only if you can participate for the entire event.

Only team submissions will be accepted. Individual submissions will not be considered. Teams that bring together students from across the University are strongly encouraged. Complete the official Google Form by **11:59 PM EST** on **Sunday, October 11, 2026**. Late submissions will not be considered.

Questions? Watch the Information Session Recording

Dr. Steven Goss led an information session to provide prospective applicants with an overview of the program, application process, and event expectations, and to answer questions before applying. If you missed the live session, you can watch the recording below to learn more before submitting your application.

- Zoom Recording

For questions, please contact Dr. Steven Goss at swg3@nyu.edu.